

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: df = sns.load_dataset('titanic')
print(df.head())
print("\nDataset Info:\n")
print(df.info())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Dataset Info:

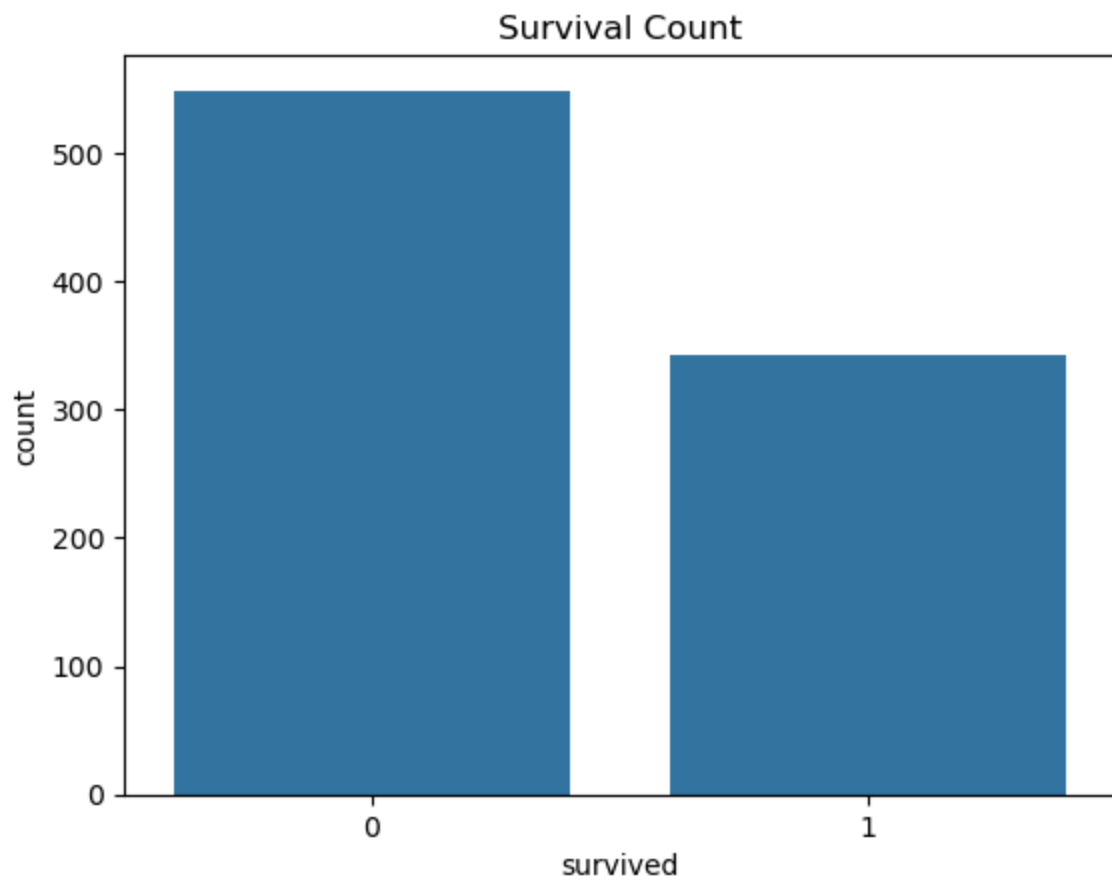
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null    int64
1   pclass          891 non-null    int64
2   sex             891 non-null    object
3   age            714 non-null    float64
4   sibsp          891 non-null    int64
5   parch          891 non-null    int64
6   fare           891 non-null    float64
7   embarked       889 non-null    object
8   class          891 non-null    category
9   who            891 non-null    object
10  adult_male     891 non-null    bool
11  deck          203 non-null    category
12  embark_town    889 non-null    object
13  alive         891 non-null    object
14  alone         891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

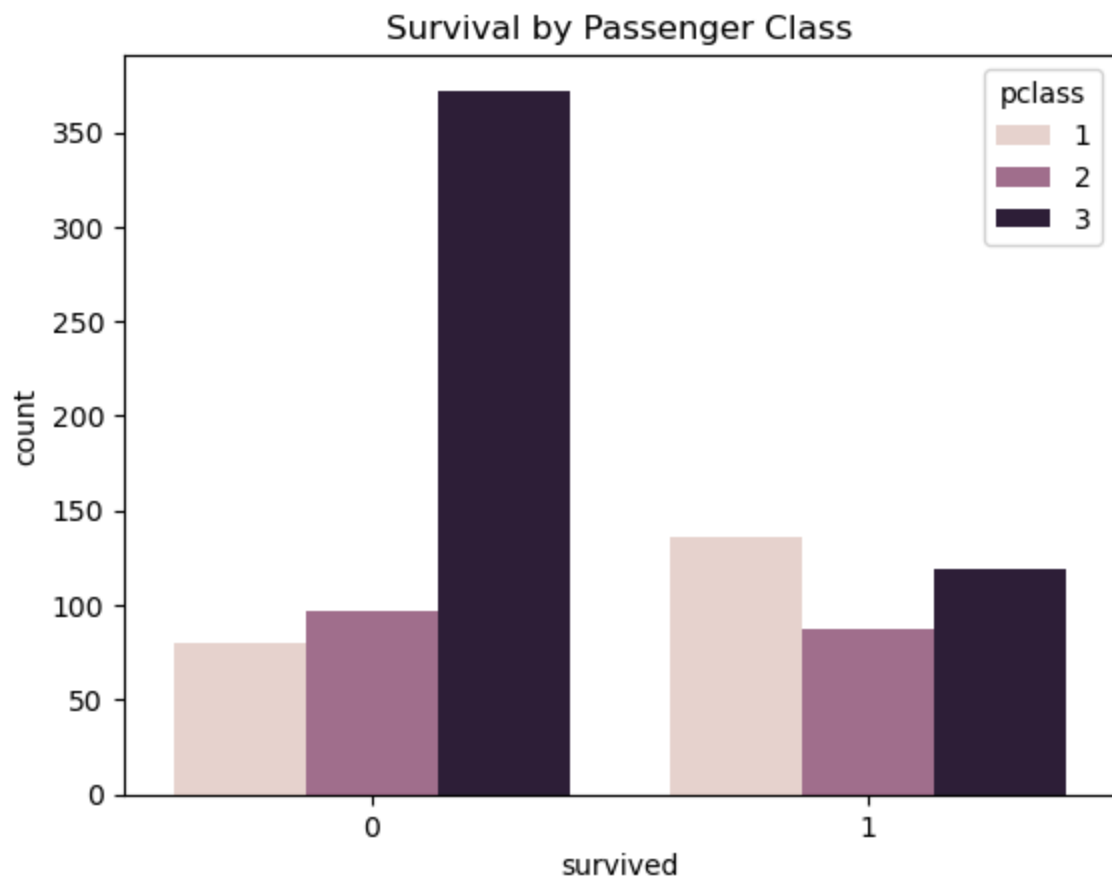
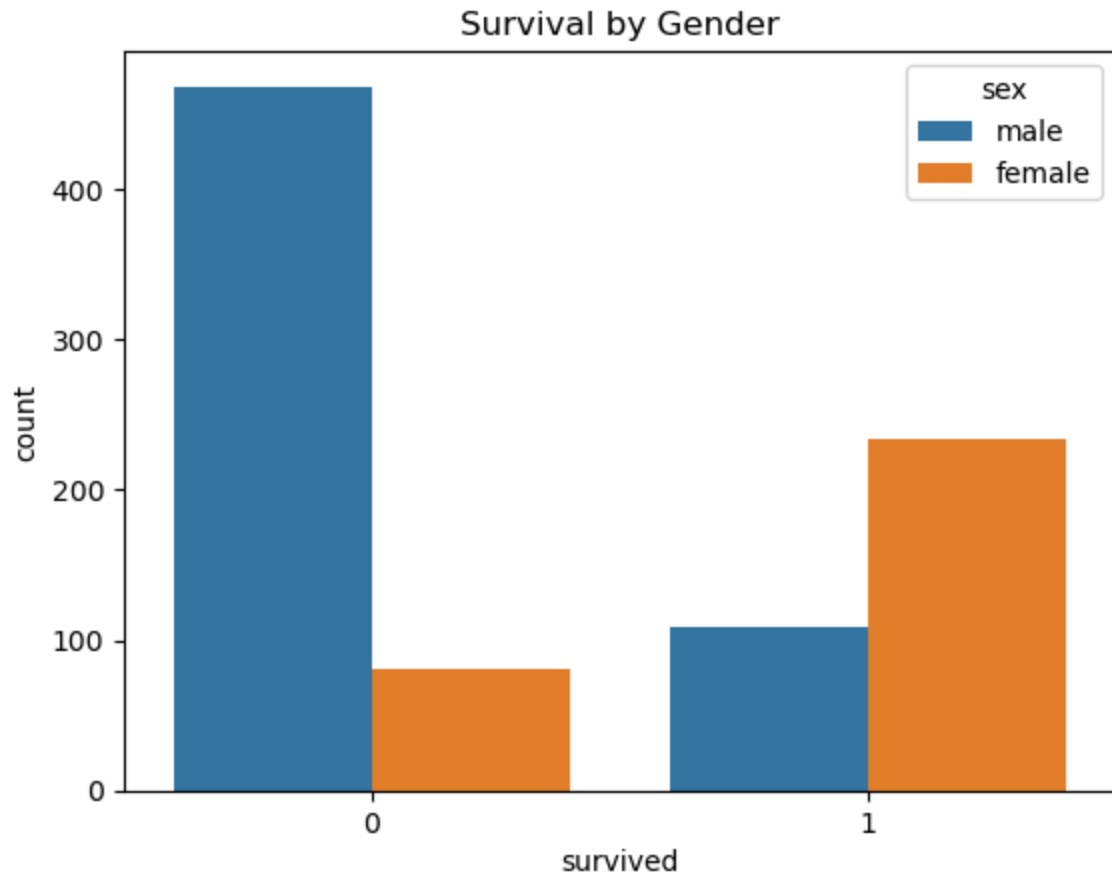
```
In [5]: # Survival count
sns.countplot(x='survived', data=df)
plt.title("Survival Count")
plt.show()
```

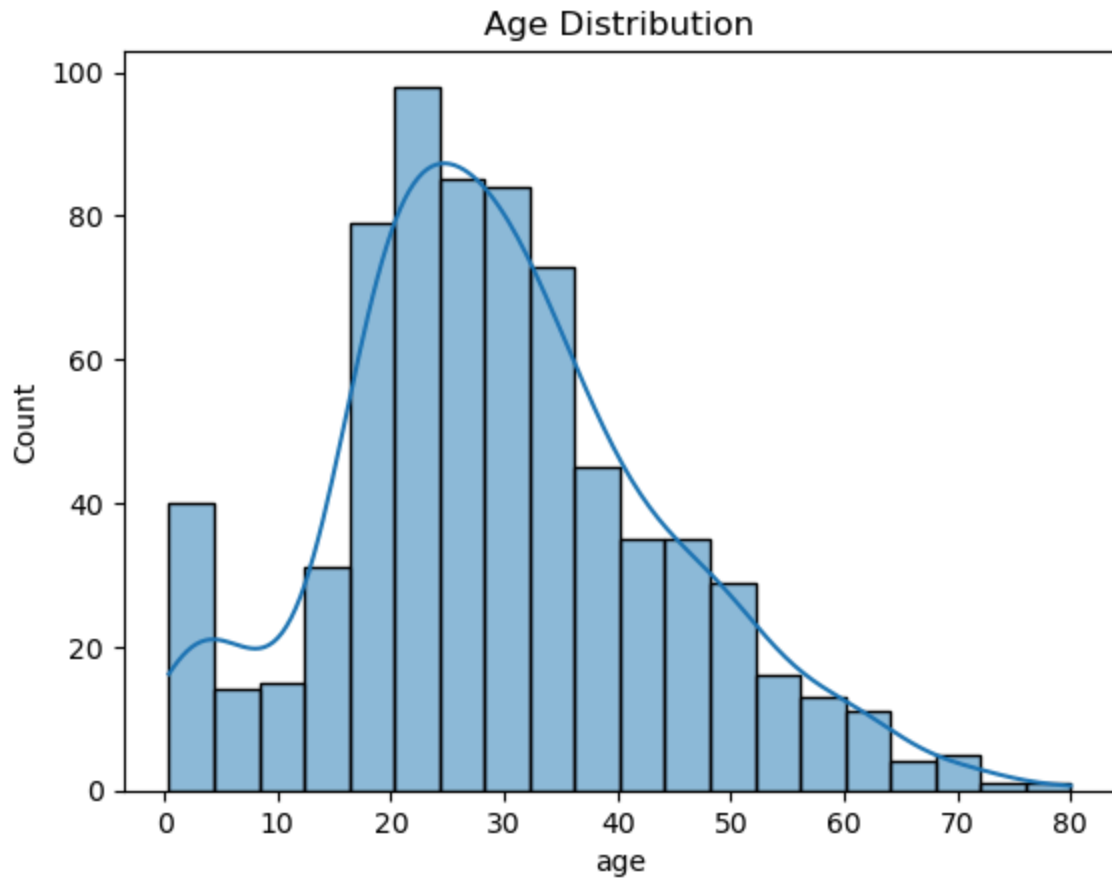
```
# Survival based on gender
sns.countplot(x='survived', hue='sex', data=df)
plt.title("Survival by Gender")
plt.show()

# Survival based on passenger class
sns.countplot(x='survived', hue='pclass', data=df)
plt.title("Survival by Passenger Class")
plt.show()

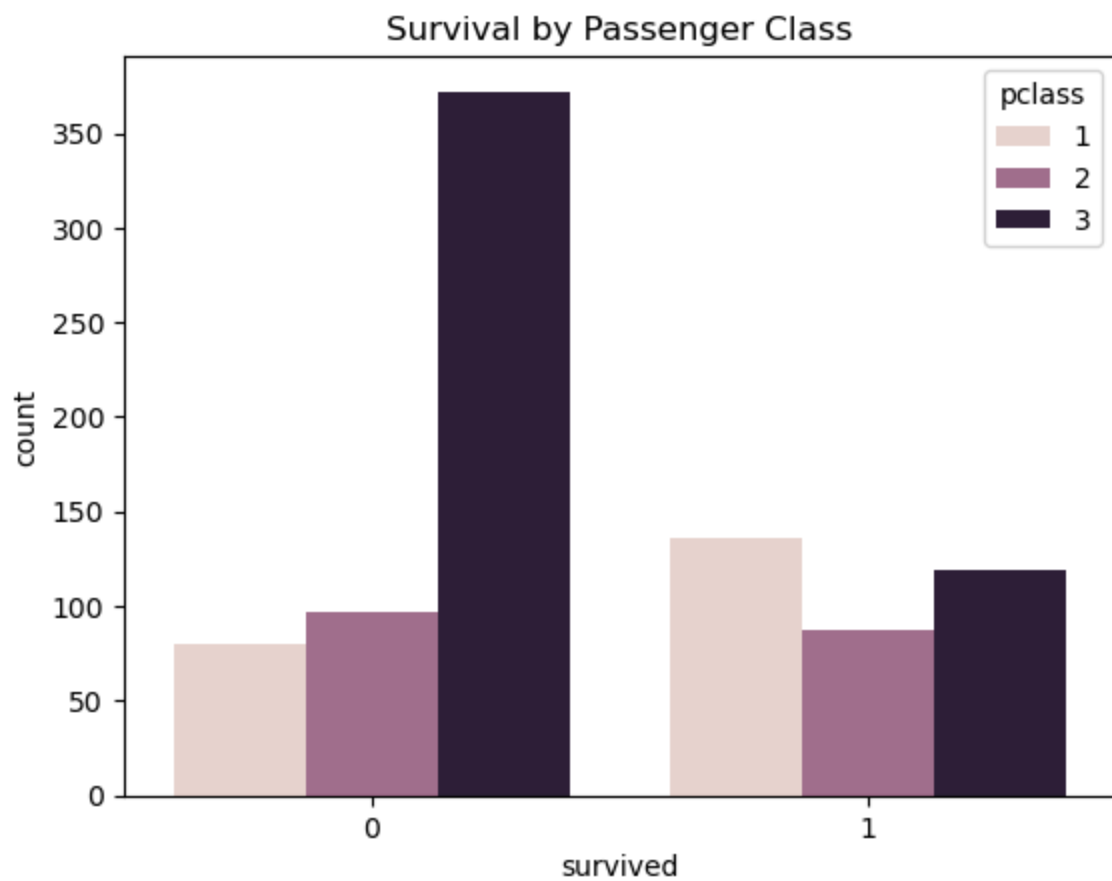
# Age distribution
sns.histplot(df['age'], kde=True)
plt.title("Age Distribution")
plt.show()
```

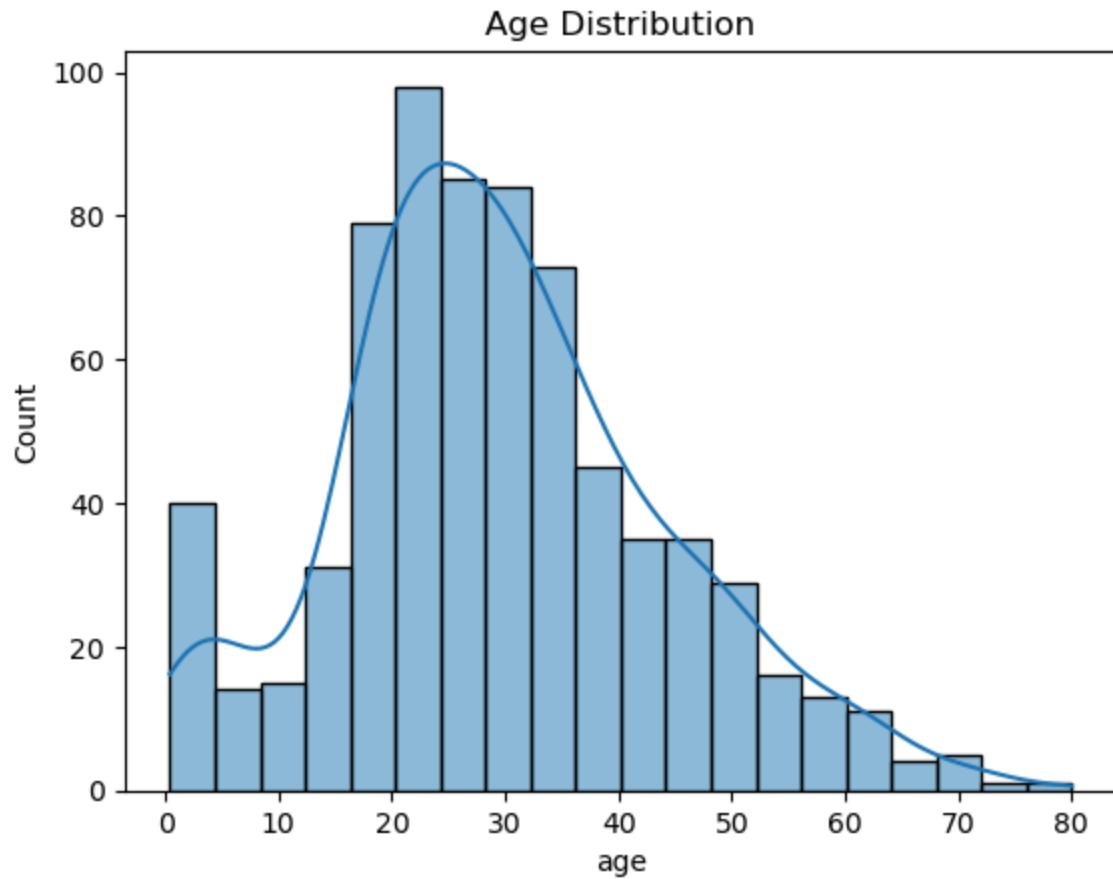




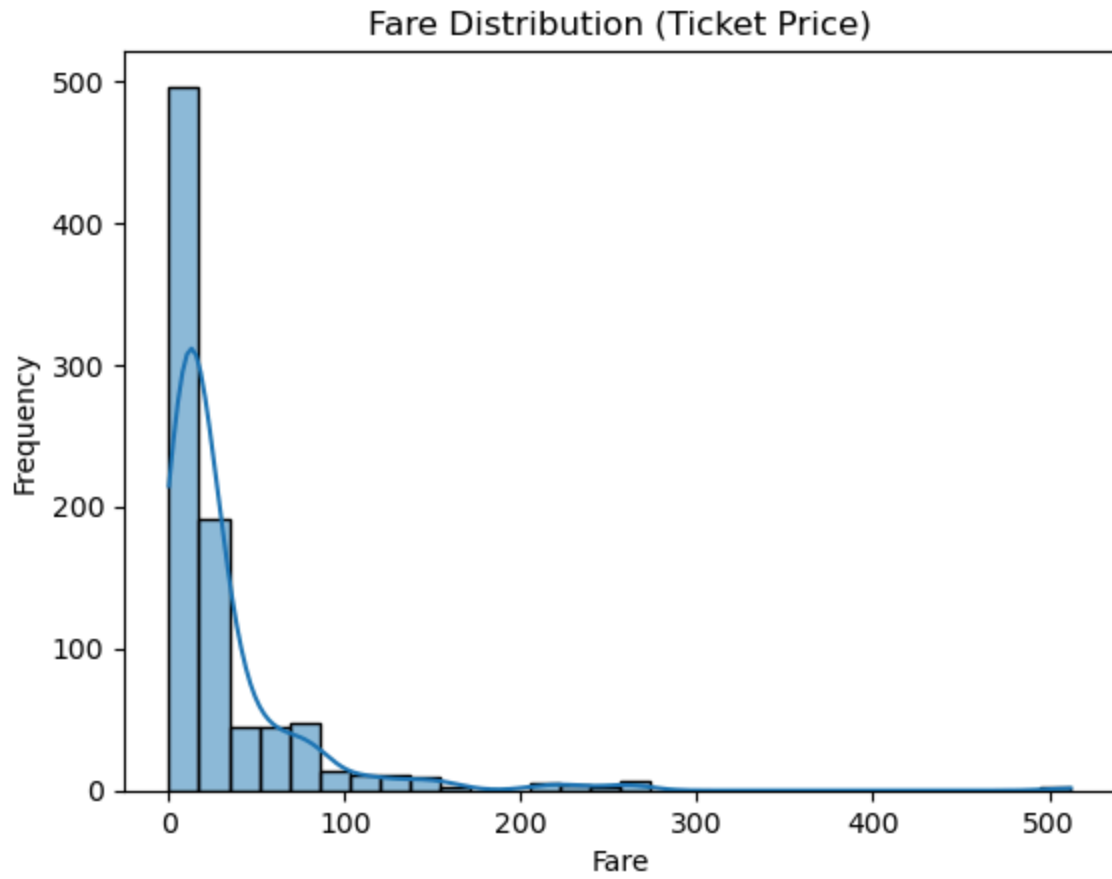


In [4]:





```
In [6]: # 3. Fare Distribution
sns.histplot(df['fare'], bins=30, kde=True)
plt.title("Fare Distribution (Ticket Price)")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



In []: